

Evaluation of

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Executive Summary

This report evaluates the risk of a bridge collapsing in Omaha, Nebraska over the next five years. The goal is to help city decision makers understand current risks and identify actions that can reduce those risks.

Data from the National Bridge Inventory and federal traffic statistics were used to analyze bridge condition, age, and traffic levels. Information from the Nebraska Department of Transportation was used to provide additional context on maintenance. Risk was defined as a combination of the probability of failure and the consequences if a failure occurs.

The analysis shows that most bridges in Omaha are currently in fair to good condition. Out of 511 bridges, only 16 have a condition rating of 4 or below, meaning they may be at higher risk. Older bridges were more likely to have lower condition ratings, suggesting that deterioration over time is an important factor.

Traffic data shows that some bridges carry significantly higher volumes of vehicles than others. If one of these high traffic bridges were to fail, it would cause major disruptions and potential safety issues. Bridges with both low condition ratings and high traffic levels represent the greatest overall risk.

Overall, the risk of widespread bridge failure in Omaha is low. However, a small number of bridges require additional attention to prevent future problems.

Recommendations

- Increase monitoring of older bridges with low condition ratings
- Prioritize repairs for bridges in poorer condition
- Continue regular inspections and maintenance
- Focus on high-traffic bridges where failure would have the greatest impact

Taking these steps will help reduce the chance of bridge failure and limit the impact if a failure occurs.

1 Introduction

Bridge safety has become more of a concern in the United States after several bridge collapses in recent years. One example is the 2022 Pittsburgh bridge collapse (*Maintenance and Oversight Failures Led to 2022 Pittsburgh Bridge Collapse*, n.d.). Events like this have caused cities to pay more attention to the condition and safety of their bridges.

In Omaha, bridges are important for everyday travel, especially on major roads like I-80 and routes that cross the Missouri River. If a bridge were to fail, it could lead to serious safety issues and major traffic problems.

The goal of this report is to estimate the risk of bridge collapse in Omaha over the next five years. This is done by analyzing bridge condition, age, and traffic levels using publicly available data. The results are meant to help the city identify higher risk bridges and understand what actions can be taken to reduce that risk.

2 Methods

This analysis primarily uses data from the National Bridge Inventory (*Download NBI ASCII Files - National Bridge Inventory - Bridge Inspection - Safety Inspection - Bridges & Structures - Federal Highway Administration*, n.d.) and Federal Highway Administration traffic data (*Highway Statistics 2023 - Policy | Federal Highway Administration*, n.d.). Additional information from Nebraska Department of Transportation reports (*Bridge - NDOT*, n.d.) is used to provide context for maintenance and repair decisions.

Only bridges located in Douglas County, Nebraska, which includes Omaha, were included in the analysis. Bridges with missing structural condition ratings were excluded to ensure consistent results.

2.1 Data

2.1.1 National Bridge Inventory

The data comes from the Federal Highway Administration’s National Bridge Inventory (*Download NBI ASCII Files - National Bridge Inventory - Bridge Inspection - Safety Inspection - Bridges & Structures - Federal Highway Administration*, n.d.). This dataset provides information about bridge age, condition ratings, location, and traffic levels. The main variables used from this dataset are shown in Table 2.1.

Table 2.1: Variables used from the National Bridge Inventory dataset.

Variable	Data_Type	Description
Year Built	Numeric	Used to calculate bridge age
Deck Rating	Ordinal (0–9)	Condition of the bridge deck
Superstructure Rating	Ordinal (0–9)	Condition of load-bearing structure
Substructure Rating	Ordinal (0–9)	Condition of bridge supports
Lowest Rating	Ordinal (0–9)	Lowest condition rating among major bridge parts
Bridge Condition	Categorical	Overall condition category such as Good, Fair, or Poor
Average Daily Traffic	Numeric	Vehicles per day

These variables help show the condition of each bridge. Lower condition ratings indicate a greater chance of structural problems. Bridge age is also an important factor because older bridges are more likely to show wear over time.

There is a possible issue with missing data in the dataset. Some bridges may not have complete ratings for every condition measure. If data is missing, those bridges will either be removed from the analysis or reviewed carefully so the results are not misleading.

Nebraska Department of Transportation

These state bridge reports are from NDOT (*Bridge - NDOT*, n.d.). This source adds more local information about repairs and maintenance plans, which can help when looking at short term risk. The main variables used from this source are shown in Table 2.2.

Table 2.2: Variables used from Nebraska Department of Transportation bridge data.

Variable	Data_Type	Description
Planned Repairs	Categorical (Yes/No)	Shows whether repairs are scheduled
Funding Levels	Numeric	Shows maintenance investment

This source is useful because a bridge with planned repairs or available funding may have lower short term risk than a bridge with no repair plans. It also helps add more local context that may not be fully shown in the national bridge inventory data.

2.1.2 Traffic Data

This is traffic data that comes from FHWA Highway Statistics. (*Highway Statistics 2023 - Policy / Federal Highway Administration*, n.d.). This source is mainly used to help estimate the consequences of a bridge collapse instead of the probability of one actually happening. The main variables used from this source are shown in Table 2.3.

2.1.2.1 Important Variables

Table 2.3: Variables used from FHWA traffic data.

Variable	Data_Type	Description
Average Annual Daily Traffic (AADT)	Numeric	Vehicles per day
Truck Percentage	Numeric (%)	Percentage of trucks in traffic

Table 2.3 helps show how serious the consequences would be if a bridge collapsed. A bridge that carries a high amount of traffic, especially truck traffic, would more likely cause more problems.

2.1.3 Is this enough data?

I do think it is enough data. Each source provides me with different things that could help assess and prevent the bridge collapse risk in Omaha over the next five years. The NBI data provides the probability of a bridge collapsing. The NDOT data helps with local repair information, and the traffic data measure the impact. I think they work together very well.

2.2 Selection of Relevant Variables

The variables used in this analysis are shown in Table 2.1, Table 2.2, and Table 2.3. The main variables include bridge age, deck rating, superstructure rating, substructure rating, lowest condition rating, bridge condition, and average daily traffic.

Bridge age was calculated using the year built. Bridge condition was measured using deck, superstructure, and substructure ratings. A bridge was classified as being in poorer condition if its lowest condition rating was 4 or below.

2.3 Evaluating Risk

To estimate the probability of failure, bridges with lower condition ratings were treated as higher risk. To estimate the consequences of failure, average daily traffic was used as the measure of impact, with higher traffic indicating greater potential disruption.

Overall bridge risk was estimated using the formula:

$$\text{Risk} = \text{Probability of Failure} \times \text{Consequences of Failure}$$

This approach was used to identify bridges that may require additional monitoring or maintenance, as discussed in Chapter 3.

3 Results

3.1 Data

The data set includes 511 bridges in the Omaha (Douglas County) area. The average bridge age is about 45 years, showing a mix of older and newer structures.

Most bridges have condition ratings that fall within fair to good categories.

Out of the 511 bridges analyzed, 16 bridges have a condition rating of 4 or below, indicating poorer condition. This represents about 3% of all bridges, suggesting that most bridges are not currently in poor condition.

Traffic levels vary across bridges. Some carry only a small number of vehicles per day, while others carry over 100,000 vehicles per day, especially on major highways.

3.2 Probability of Failure

The analysis shows that bridges with lower condition ratings have a higher risk of failure. Bridges with a lowest condition rating of 4 or below were identified as higher risk bridges.

The results also show that older bridges tend to have lower condition ratings. This suggests that bridge condition declines over time due to wear and usage. While most bridges are still in acceptable condition, there is a small group of bridges with lower ratings that may be at higher risk.

3.3 Consequences of Failure

Traffic data shows that bridges with higher daily traffic would cause the largest disruptions if they failed. Bridges that carry very high traffic volumes, especially on major roads like I-80, would have the greatest impact on transportation and public safety.

Information from NDOT (*Bridge - NDOT*, n.d.) provides context on maintenance and planned repairs, which may help reduce short-term risk. However, bridges that have both high traffic and low condition ratings represent the highest overall risk.

These higher risk bridges are typically located along major transportation routes in Omaha. Since these routes are heavily used, a failure in these areas would cause major traffic disruptions and safety concerns.

{=html}

4 Conclusion

The results show that most bridges in Omaha are in fair to good condition, with only 16 out of 511 bridges having a condition rating of 4 or below, which ends up being around 3%. This means the overall risk of a bridge collapse is low, but there are still some bridges that may need attention.

The data also shows that older bridges tend to have lower condition ratings, and bridges with higher traffic would cause the biggest problems if a failure occurred. Due to this, bridges that are both older and heavily used represent the highest risk.

Based on these findings, the city should focus on monitoring bridges with lower ratings more closely, prioritize repairs for these structures, and continue regular inspections. These actions would help reduce both the chance of failure and the impact if one occurs.

While more detailed data could improve these estimates in the future, focusing on maintenance and monitoring is currently the best way to reduce the risk of a bridge collapse in Omaha over the next five years.

References

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